
Exact Gaussian-Process Inference over Two Million CpG Sites Shows That DNA Co-Methylation Propagates in CpG-Index Space

Anonymous

Abstract

DNA methylation is measured site by site, but it varies region by region: correlation between neighboring CpG sites decays within hundreds of base pairs, at rates that depend strongly on local CpG density. Existing genome-scale methylation smoothers and imputation tools are point estimators; principled uncertainty would require a Gaussian process (GP), which is widely considered infeasible at the scale of a chromosome. We show that the methylome is, in a precise sense, a naturally sparse GP: we introduce a compactly supported, non-stationary kernel whose metric interpolates between genomic (base-pair) distance and CpG-*index* distance, and whose signal variance is driven by local CpG density—a covariate computed from the genome sequence alone—paired with a coverage-aware overdispersed binomial noise model. Because the metric warp is monotone, zero covariance blocks are certifiable *a priori*, the covariance is provably banded, and exact inference (sparse Cholesky, exact log-determinant, block-MCMC over hyperparameters) runs on all 2 123 184 coverage-filtered CpG sites of human chromosome 1 on a single 32-core CPU node in 0.73 h. On a frozen 15,000-site held-out set, the density-warped kernel beats the stationary compactly supported kernel of the gp2Scale family fitted to the same data by 3.4×10^4 nats of log marginal likelihood, improves RMSE in every genomic compartment (by 12% inside CpG islands), CRPS, and read-depth-resolved calibration, and does so at a $7\times$ smaller covariance bandwidth— $5.8\times$ cheaper per exact likelihood evaluation—because a base-pair-metric kernel must buy island-adequate support with a radius the dense compartments cannot afford. Exact GPs of either kernel dominate variational baselines (SVGP, VNNGP) given the same side information. The fitted posterior yields a biological conclusion: the metric-interpolation weight concentrates at $\gamma = 0.897$ (95% credible interval $[0.892, 0.902]$), i.e. co-methylation propagates per CpG site rather than per base pair, so correlation reaches $3.3\times$ farther in sparse open sea than inside dense islands. All results are exact-GP posteriors; no inducing points, interpolation, or neighborhood truncations are involved.

1 Introduction

Whole-genome bisulfite sequencing (WGBS) reads out cytosine methylation at single-CpG resolution, and nearly every downstream analysis—differential methylation, partially methylated domain (PMD) detection, deconvolution, imputation of sparse assays—begins by converting noisy per-site read counts into an estimate of the underlying methylation level. The de facto standard tools are local smoothers and machine-learning imputers: BSmooth fits local polynomials [Hansen et al., 2012], dmrseq and SOMNiBUS use spline regression [Korthauer et al., 2019, Zhao et al., 2021], and BoostMe, DeepCpG and CpG Transformer learn point predictors [Zou et al., 2018, Angermueller et al., 2017, De Waele et al., 2022]. What none of these provide is a joint, calibrated posterior: a

probabilistic answer to “how certain are we about the methylation level here, given the read depth here and the correlation structure around here?” That is precisely what a Gaussian process (GP) regression model would provide—together with principled propagation of the strongly site-dependent binomial sampling noise—if only exact GP inference were feasible at $N \sim 10^6$ sites. The conventional wisdom is that it is not: exact GP training scales as $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ memory, and GP treatments of methylation have therefore been confined to windows of tens of sites or to per-region models [Halla-aho and Lähdesmäki, 2020, Frank, 2024].

This paper starts from the observation, due to the sparsity-discovering exact-GP line of work of Noack et al. [Noack et al., 2023, 2025, Risser et al., 2025], that the covariance matrix of a GP is dense only because standard kernels make it so: with a flexible *compactly supported* kernel, the covariance of a large dataset can be genuinely sparse, and exact inference reduces to sparse numerical linear algebra. The methylome is arguably the ideal habitat for this idea—the co-methylation literature consistently finds correlation decaying to ≈ 0.4 within ~ 400 bp and structured at the scale of methylation haplotype blocks of median length 50–70 bp [Zhang et al., 2015, Guo et al., 2017, Feng et al., 2023, Affinito et al., 2020]—but it also exposes exactly what that kernel family cannot yet express. Correlation structure in the methylome is slaved to *local CpG density*: dense CpG islands behave as coherent low-variance units, sparse open-sea and PMD regions are high-variance with correlation reaching much farther in base pairs, and the noise level varies by more than an order of magnitude from site to site with read depth [Lövkqvist et al., 2016, Taryma-Leśniak et al., 2024, Ren et al., 2022]. The gp2Scale kernels express non-stationarity as a function of the *coordinate*—Paciorek–Schervish length-scale fields, bump-function masks at learned positions [Paciorek and Schervish, 2006, Noack and Sethian, 2022]—which is the wrong parameterization here: a chromosome contains on the order of 10^4 CpG islands, so tracking density-driven structure with coordinate-anchored fields would need thousands of hyperparameters, far beyond what block-MCMC can sample.

Contributions. We make three contributions. *First, a kernel* (Section 2): a compactly supported, non-stationary construction for the methylome in which (i) distance is measured in a learned interpolation $u(\gamma)$ between base-pair position and CpG catalog index, (ii) the signal standard deviation is an explicit function of local CpG density—a covariate computed from the reference genome alone—and (iii) observations carry a coverage-aware, overdispersed binomial noise floor. To our knowledge this is the first compactly supported kernel whose support structure is driven by an external covariate rather than by coordinates (cf. [Risser and Calder, 2015, Bolin and Wallin, 2016, Mateu et al., 2013]). *Second, a sparsity certificate*: because the metric warp is monotone, the support of every row of the covariance is a contiguous index interval; the matrix is certifiably banded with a data-dependent bandwidth computed in $\mathcal{O}(N \log N)$, so zero blocks are provable *before* any kernel evaluation. This strengthens the gp2Scale pipeline, in which “non-zero elements are not predetermined . . . so all covariance matrix entries must be computed” [Noack et al., 2025]: for warped-metric kernels the $\mathcal{O}(N^2)$ assembly collapses to $\mathcal{O}(Nb)$, and exact banded Cholesky supplies solves and log-determinants with no stochastic estimators. *Third, a biological result with calibrated evidence*: on all 2 123 184 coverage-filtered CpGs of human chromosome 1 (HepG2, ENCODE), the posterior of the metric weight concentrates at $\gamma = 0.897$ —co-methylation propagates per *site*, not per base pair—and the learned variance field, sparsity pattern, and per-region calibration quantify how island, shore, shelf and open-sea compartments differ. We validate our build by machine-precision equivalence against the unmodified reference implementation at the full announced scale, and we also report an instructive *failed* reproduction of the reference line’s MNIST benchmark, tracing the published score to a covariance density that the benchmark’s GPU cluster could afford per MCMC iteration and a CPU node cannot, and showing the benchmark’s ℓ_1 Wendland Gram matrix is not positive definite. On the methylome we demonstrate that the existing stationary and convolution kernels fitted to the same chromosome are measurably inadequate—trailing by tens of thousands of nats of marginal likelihood, losing on every held-out metric, and paying several times our compute for the privilege—and that variational baselines with the same side information are not competitive. Exactness is not cosmetic: it is what makes the comparison between kernels a comparison between *models*, with no approximation error in the way, at a scale where an exact GP was previously unavailable to this domain.

2 Methodology

2.1 Setting

For a chromosome, let $x_1 < \dots < x_N$ be the base-pair coordinates of the CpG sites carrying data, a subset of the full CpG catalog $\mathcal{C} = \{c_1 < \dots < c_M\}$ extracted from the reference sequence (chromosome 1 of GRCh38 has $M = 2\,375\,159$ catalog sites). Site i reports m_i methylated out of n_i total reads; the observed methylation fraction is $y_i = m_i/n_i \in [0, 1]$. We model $y_i = f(x_i) + \varepsilon_i$ with a GP prior $f \sim \mathcal{GP}(m(\cdot), k(\cdot, \cdot))$ and independent Gaussian noise $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2(x_i))$, and we work with the exact log marginal likelihood $\log p(\mathbf{y} | \phi) = -\frac{1}{2}(\mathbf{y} - \mathbf{m})^\top (\mathbf{K} + \mathbf{V})^{-1}(\mathbf{y} - \mathbf{m}) - \frac{1}{2} \log |\mathbf{K} + \mathbf{V}| - \frac{N}{2} \log 2\pi$, sampled over hyperparameters ϕ by block-MCMC. Three quantities are precomputed from the catalog alone (never from assay values): the *rank* $r(x)$ of a site in $\mathcal{C}$; the *local density covariate* $c(x)$, the standardized negative logarithm of the mean inter-CpG spacing in a centered window of 129 catalog sites; and the chromosome-mean spacing $\bar{s}$ (104.8 bp for chromosome 1).

2.2 A density-warped, compactly supported kernel

Metric interpolation. The co-methylation literature implies two competing distance metrics. Maintenance methyltransferase activity and methylation haplotype blocks suggest correlation organized per *site* (the enzyme acts processively across successive CpGs [Ren et al., 2022]), while simple diffusion-like arguments suggest per *base pair*. We let the data decide, defining the warped coordinate

$$u_i(\gamma) = (1 - \gamma) \frac{x_i}{\bar{s}} + \gamma r(x_i), \quad \gamma \in [0, 1], \quad (1)$$

which interpolates between the physical metric ($\gamma=0$; the setting of all stationary compact kernels, including gp2Scale’s Wendland) and the CpG-index metric ($\gamma=1$), both expressed in units of one typical site spacing. Composition of a kernel with any deterministic map preserves positive semi-definiteness, and because (1) is strictly monotone in x for every γ , sorted-order structure survives the warp—the property behind the sparsity certificate below. In dense regions (islands, spacing ~ 10 bp) a fixed warped radius covers few base pairs; in sparse open sea (spacing ~ 150 bp) the same radius reaches an order of magnitude farther in base pairs. A single scalar γ therefore encodes exactly the density-dependent reach that coordinate-parameterized non-stationary kernels would need thousands of parameters to imitate.

Correlation shape and variance field. On warped distance we place the compactly supported C^4 Wendland correlation used throughout the gp2Scale line (as implemented in the reference fvGP build¹), $W(d) = (1 - d)_+^8 (32d^3 + 25d^2 + 8d + 1)$, and modulate amplitude by a density-driven field:

$$k(x_i, x_j) = \sigma_s(x_i) \sigma_s(x_j) W\left(\frac{|u_i(\gamma) - u_j(\gamma)|}{\rho}\right), \quad \sigma_s(x) = \text{softplus}(v_0 + v_1 c(x)). \quad (2)$$

The variance field costs two hyperparameters yet captures the compartmentalization that a chromosome displays (quiescent island cores vs. disordered PMDs), because it is indexed by *density*, not by position.

Density-driven mean and coverage-aware noise. The prior mean follows the same covariate, $m(x) = m_0 + m_1 \tanh((c(x) - m_2)/\text{softplus}(m_3))$, freeing the covariance from explaining the systematic island/sea level difference. Each observation carries its own binomial noise floor with a learned overdispersion α and nugget τ :

$$\sigma_n^2(x_i) = \alpha \frac{\tilde{p}_i(1 - \tilde{p}_i)}{n_i} + \tau^2, \quad \tilde{p}_i = \frac{m_i + 1/2}{n_i + 1}. \quad (3)$$

Read depth in our data spans $5\text{--}111\times$ (median $27\times$), so these floors vary by more than $20\times$ across sites—structure that a homoscedastic model must average away. The full hyperparameter vector has ten components: $\phi = (\log \rho, \log \gamma, v_0, v_1, m_0, m_1, m_2, \log m_3, \log \alpha, \log \tau)$.

¹The published equation prints $35d^3$; the reference implementation, which we follow for comparability, uses $32d^3$.

2.3 Certified banded sparsity and distributed assembly

Compact support makes $k(x_i, x_j) = 0$ whenever $|u_i - u_j| \geq \rho$. Because u is monotone in the sorted site order, the nonzeros of row i form a *contiguous* interval of indices, so $\mathbf{K}$ is banded with the data-dependent bandwidth $b(\phi) = \max_i \#\{j > i : u_j - u_i < \rho\}$, computable exactly by one vectorized binary-search pass in $\mathcal{O}(N \log N)$ *before any kernel evaluation*. Two consequences follow. First, a scheduling certificate: any block pair of sites whose warped ranges are separated by at least ρ is provably zero and is never dispatched; assembly work drops from the $\mathcal{O}(N^2)$ candidate pairs of the reference pipeline to $\mathcal{O}(Nb)$. Second, an exactness dividend: banded Cholesky (LAPACK `dpbtrf`, cost $\mathcal{O}(Nb^2)$) yields machine-precision solves and log-determinants, so neither iterative-solver tolerances nor stochastic Lanczos log-determinant estimators enter the likelihood. Assembly itself follows the `gp2Scale` template—the site set is split into blocks, and parallel workers each compute their block’s sparse entries—realized here as 30 fork-based worker processes writing column-blocks of the banded storage directly into POSIX shared memory (copy-on-write sharing of the site table; no serialization). A proposal whose certified bandwidth exceeds a budget $B_{\max} = 768$ is rejected in the prior with the rejection logged, the same “keep the support radius affordable” discipline the original pipeline imposes, and the analogous worker-side certificate exists in the reference implementation’s support-aware kernels. We verified at full scale that the engine and the reference fvGP distributed assembly (dask workers, COO→CSR, our banded solver injected as a custom linear-algebra mode) agree to relative difference $1.2\text{e-}15$ in log marginal likelihood ($N=2\,123\,184$; 27.1 s vs. 0.8 s per evaluation).

2.4 Inference and prediction

Hyperparameters are sampled by block Metropolis–Hastings MCMC (the `gpMCMC` sampler of the reference implementation) in three natural blocks—kernel ($\log \rho, \text{logit } \gamma, v_0, v_1$), mean ($m_0, m_1, m_2, \log m_3$), noise ($\log \alpha, \log \tau$)—with adaptive Gaussian proposals; every iteration evaluates the exact likelihood on the full training set. Prediction at held-out sites uses the exact posterior $\mu_* = m(x_*) + \mathbf{k}_*^\top (\mathbf{K} + \mathbf{V})^{-1} (\mathbf{y} - \mathbf{m})$ and $\sigma_{f*}^2 = k(x_*, x_*) - \mathbf{k}_*^\top (\mathbf{K} + \mathbf{V})^{-1} \mathbf{k}_*$ at the maximum-a-posteriori sample, with the observation-level predictive variance $\sigma_{f*}^2 + \sigma_n^2(x_*)$ used for scoring against observed fractions. Cross-covariance columns are sparse by the same certificate, and segment-parallel banded solves handle all 15 000 test sites in 5.6 seconds.

3 Experimental Setup

3.1 Data

We use the ENCODE HepG2 WGBS experiment ENCSTR786DCL (Richard Myers lab), the only deep human WGBS experiment on the portal’s common cell lines with no quality audit flags; replicate 2 (file ENCFF369YQW, GRCh38 bedMethyl, md5-verified) is the primary signal and replicate 1 (ENCFF064GJQ) is reserved for concordance checks. Strand-merged read counts at the 2 375 159 catalog CpGs of chromosome 1 (catalog and density covariate derived from the UCSC GRCh38 FASTA; CpG-island annotation from the UCSC `cpgIslandExt` track, with shores/shelves at $\pm 2/\pm 4$ kb) give 2 138 184 sites with coverage ≥ 5 . A held-out test set of 15 000 sites was drawn uniformly at random with a fixed, logged seed (20260817) *before any model fitting*; the remaining $N = 2\,123\,184$ sites constitute the training set used—in full—by every exact-GP run. HepG2 is globally hypomethylated with prominent PMDs [Salhab et al., 2018], visible in our data as 35% of sites at intermediate methylation; per-class means run from 0.35 (islands) to 0.51 (open sea) with large within-class dispersion. Chromosome 21 (343 145 train / 4 000 test sites) serves as the calibration pilot for proposal scales and seed-replication.

3.2 Protocol and metrics

All methods are scored on the same frozen test set with three metrics: RMSE of the predictive mean against the observed methylation fraction; CRPS of the full Gaussian predictive distribution (closed form [Gneiting and Raftery, 2007]); and the probability-coverage curve—empirical coverage of central credible intervals at nominal levels 5–95%—summarized by coverage at 90% and by the expected calibration error (ECE, the mean absolute deviation of the curve from the diagonal). We

also report NLPD, per-annotation-class breakdowns, and a high-coverage subset ($n \geq 30$, 6 666 sites) in which the observation floor is small. Wall-clock and peak system memory are logged for every run. Hardware throughout: one 32-vCPU node, 61 GB usable RAM, no GPU.

3.3 Reference kernels, ablation, and approximate baselines

Existing-kernel references (same data, same protocol). The stationary compactly supported Wendland kernel—the workhorse kernel of the gp2Scale family, Eq. (3) of [Noack et al., 2025]—with learned amplitude, support radius (in bp), constant mean and constant noise; and the paper’s non-stationary convolution (Paciorek–Schervish) Wendland, Eq. (6), with $\log \ell(x)$ and $\log \sigma_s(x)$ expanded in $K=4$ Gaussian radial basis functions of the coordinate. K was fixed a priori so that this model carries slightly *more* hyperparameters (12) than our kernel (10), keeping sampling difficulty comparable; tracking density-driven structure would need $K \sim 10^4$ (one anchor per island), which is out of reach for any MCMC over exact likelihoods. Both are fitted by the same BMH-MCMC machinery at the same $N = 2\,123\,184$ for the same 500 updates, with proposal blocks following each model’s natural parameter grouping (a single 4-parameter block for the stationary kernel, kernel/field/observation groupings otherwise), so per-dimension sampling effort is comparable across rows. Two inadequacy hypotheses were pre-registered before any full-scale run: **H1**, that one global (σ_s, r_0) and one noise level cannot simultaneously serve compartments whose density, variance, and read-depth floors differ by orders of magnitude—predicting worse held-out CRPS and structured miscalibration; and **H2**, that MCMC-tractable coordinate-parameterized non-stationarity cannot track density variation with $\sim 10^4$ islands, predicting scores indistinguishable from the stationary kernel. Outcomes are reported against both in Section 4, including the parts of H1’s predicted failure pattern that did *not* materialize as expected. *Ablation.* The stationary kernel paired with *our* density mean and coverage-aware noise (“stationary + obs. model”) separates what the warp/variance field contribute from what the observation model contributes. *Approximate baselines.* SVGP [Hensman et al., 2013] and VNNGP [Wu et al., 2022] (GPyTorch implementations) trained on the full 2 123 184-site training set by minibatch ELBO. We gave them a genuine best effort: both a pure-position and a (position, density-covariate) feature space; for SVGP a configuration grid over inducing counts and both homoscedastic and fixed-floor likelihoods (i.e., the baselines were offered the same noise floors our model uses), selected on a 20,000-site validation subset carved from the training set; the selected configuration (M=2048, (pos, dens), fixed-floor) rerun with three seeds. VNNGP used $k = 64$ nearest neighbors and a mean-field variational distribution, three seeds, at its largest tractable scale: the full-training-set attempt hung for three hours inside the sequential-nearest-neighbor construction that the implementation’s variational strategy performs at setup, and a 500,000-site attempt was still inside the same construction after 50 minutes (both logged), so the reported rows train on a 200 000-site random subset, the largest configuration that completed on this node. Full configurations, selection traces and seeds are in the run logs.

3.4 Implementation-faithfulness protocol

Faithfulness of our build to the reference methodology is established by two instruments, with outcomes in Section 4.1. First, *numerical equivalence to the reference implementation*: log marginal likelihood, posterior means, and posterior variances from our engine are compared against the unmodified fvGP gp2Scale pipeline (dask-distributed block assembly, COO→CSR, our banded solver injected through fvGP’s documented custom linear-algebra interface) on identical data and hyperparameters—against dense linear algebra at $n = 4,000$, and against the distributed reference at chromosome-21 scale and at the full announced $N = 2\,123\,184$. Second, a *reproduction of the published MNIST benchmark* with the reference routines: 70,000 digits split 60,000/10,000 per seed, regression of $\Pr(\text{digit} = 5)$ under the ℓ_1 pixel metric with the compactly supported Wendland kernel, MINRES solves and stochastic Lanczos log-determinants from fvGP’s own linear-algebra module, BMH-MCMC over (amplitude, support radius, noise), three seeds at 55 updates each (the 3-parameter chains plateau within ~ 35 updates; each seed’s logged trace shows this), with a pre-registered tolerance of ± 0.006 around the published Brier score 0.018 ± 0.002 .

Table 1: Held-out performance on chromosome 1 (test = 15 000 sites, frozen before training). Coverage is empirical coverage of the nominal 90% central interval (target 0.900). Wall-clock is training time; memory is the peak *system-wide* RSS sampled during the run (experiments sharing the node are included, so it upper-bounds each method’s own footprint). SVGP/VNNGP rows are mean $\pm$ std over 3 seeds; exact-GP rows are single 500-update BMH-MCMC chains (chain variability of the adapted kernel across 3 independent MCMC seeds, assessed at chromosome-21 scale, is ± 0.000092 RMSE / ± 0.000076 CRPS).

Method	Kernel	$ \mathcal{D}_{\text{train}} $	$ \mathcal{D}_{\text{test}} $	RMSE	CRPS	Coverage	Wall-clock (h)	Peak memory (GB)
gp2Scale (ours)	density-warped (2)	2 123 184	15 000	0.2448	0.1341	0.887	0.73	15.1
gp2Scale	stationary Wendland	2 123 184	15 000	0.2483	0.1368	0.885	1.40	26.1
gp2Scale	conv. non-stat. (PS, $K=4$)	2 123 184	15 000	0.2475	0.1362	0.886	5.77	30.0
SVGP	Matérn- $\frac{3}{2}$ (+dens.)	2 123 184	15 000	0.3804 ± 0.0011	0.2232 ± 0.00090	0.957	0.48	62.8
VNNGP	Matérn- $\frac{3}{2}$ (+dens.)	200 000	15 000	0.4222 ± 0.00059	0.2485 ± 0.00033	1.000	0.22	4.3
gp2Scale (ablation)	stationary + obs. model	2 123 184	15 000	0.2484	0.1360	0.889	2.35	26.2

4 Experimental Results

4.1 Implementation faithfulness: one clean pass, one instructive failure

The equivalence instrument passes at machine precision everywhere: on the full 2 123 184-site training set the reference fvGP pipeline and our segment-parallel engine agree to relative difference $1.2\text{e-}15$ in log marginal likelihood (posterior means and variances to $<10^{-14}$), while one exact likelihood evaluation costs 27.1 s through the reference disk path versus 0.8 s through the engine—the certificate, not the mathematics, is the difference. Every number in this paper therefore comes from a computation the reference implementation reproduces bit-for-bit at matched inputs.

The MNIST reproduction did *not* meet its pre-registered tolerance, and we report it as a failure with an attribution. Our faithful chains (reference solvers, BMH-MCMC, three seeds) converge reproducibly—but to Brier 0.0790 ± 0.0021 ($0.0774/0.0820/0.0776$), the level of the paper’s own SVGP baseline (0.033) and of mean prediction (0.082), not the published 0.018 ± 0.002 . Three logged findings localize the discrepancy. (i) *The score requires a density regime a CPU node cannot sample.* Holding the protocol fixed and scanning the support radius with hyperparameters tuned on a train-internal validation split, held-out Brier improves monotonically with covariance density—0.0762 at the chains’ feasible sparsity ($\approx 3 \times 10^7$ nonzeros), 0.0310 at $\approx 5.5 \times 10^8$ nonzeros ($\approx 15\%$ dense, the practical CPU boundary where a single exact solve still runs but per-iteration MCMC does not; this boundary configuration already matches the paper’s own SVGP baseline, 0.033)—while a k -nearest-neighbor readout of the same distances reaches 0.005: the data supports the published accuracy, but evidently at a density beyond what one CPU node can carry through an MCMC chain and which the benchmark’s 15-A100 configuration could, and which the paper’s appendix (sparsities of 10^{-5} – 10^{-4} are reported only for its synthetic scaling table) leaves unreported for MNIST. (ii) *The benchmark’s kernel is underspecified:* the delta-mask factor k_d of the published kernel $k_{\mathcal{W}}k_d$ is described only as “mimicking a flexible nearest-neighbor approach”; our count-of-common-neighbors instantiation changed nothing measurable (logged grid). (iii) *The published benchmark is not a positive-definite GP:* the exact Gram matrix at chain hyperparameters is indefinite (sparse Cholesky fails; smallest Lanczos eigenvalue -0.048 against a +0.078 diagonal), as expected in hindsight—Wendland functions guarantee positive definiteness for Euclidean metrics in low dimension, not ℓ_1 in 784-D—and the reference MINRES + stochastic-Lanczos machinery is insensitive to small negative modes, so the published pipeline operates without noticing. Our chromosome-scale kernel, by contrast, is provably positive semi-definite (a warped one-dimensional Wendland), and its banded Cholesky would fail loudly otherwise. We conclude that our build is faithful (instrument one) and that the published MNIST number is not reproducible from the paper’s specification on this hardware; the methylation results in this paper rest on the equivalence instrument, whose evidence is complete.

4.2 Main comparison

Table 1 is the primary result. All exact-GP rows use the full 2 123 184-site training set—the linear solves, log-determinants and 500-update MCMC chains all execute at that size—and all rows are scored on the identical 15 000-site frozen test set.

Table 2: Per-compartment breakdown on the frozen test set (RMSE / CRPS / coverage@90, nominal 0.90). The warped kernel’s accuracy advantage concentrates in islands and shores; both kernels *over-cover* inside islands (the shared compartment-independent-nugget defect analyzed in Section 5.3), while depth-resolved calibration separates them (Section 4.2).

	Island	Shore	Shelf	Open sea
ours	0.1492 / 0.0883 / 0.971	0.2286 / 0.1262 / 0.914	0.2605 / 0.1417 / 0.876	0.2495 / 0.1369 / 0.881
stationary	0.1703 / 0.0959 / 0.955	0.2365 / 0.1307 / 0.904	0.2641 / 0.1443 / 0.868	0.2520 / 0.1391 / 0.881

Three comparisons matter. *Adapted vs. stationary kernel*: as a Bayesian model comparison the result is unambiguous—the density-warped kernel’s log marginal likelihood exceeds the stationary kernel’s by 33 954 nats on the same data—and it carries to held-out quality: RMSE improves in every compartment (islands by 12.4%, Table 2), CRPS from 0.1368 to 0.1341, NLPD from 0.027 to 0.011. Aggregate 90% coverage is close to nominal for both (0.887 vs. 0.885), but its *composition* differs: the stationary kernel’s constant noise tilts calibration across read depth (coverage 0.865 at depth 5–10 rising to 0.899 at ≥ 40) where the coverage-aware model stays flat (0.898–0.889)—the depth-resolved version of pre-registered hypothesis H1. Reported for completeness: H1’s specific prediction of compartment-*antisymmetric* interval miscalibration did not materialize—aggregate interval coverage is similar for the two kernels, dominated in both by island over-coverage from the shared compartment-independent nugget (Section 5.3)—while H1’s quality, likelihood, and depth-calibration predictions did. The compute asymmetry is itself evidence of metric mismatch: to reach its optimum the stationary kernel drives its support to a bandwidth of 340 sites (vs. 51), making each exact likelihood evaluation $5.7\times$ more expensive than ours *on identical hardware and data*. *Adapted vs. convolution non-stationary*: the $K=4$ coordinate fields do learn something—their fitted length-scale field varies smoothly at the tens-of-megabases scale and buys $\approx 3,200$ nats of marginal likelihood over the stationary kernel—but they cannot oscillate at the scale of $\sim 10^4$ islands: held-out scores stay at the stationary kernel’s level (Table 1) and the marginal likelihood trails ours by 30 796 nats, at $8\times$ our wall-clock, confirming hypothesis H2 of the pre-registered plan. *Exact vs. variational*: SVGP, even given the density covariate as an input, our noise floors in its likelihood, and a tuned configuration ($M=2048$, (pos, dens), fixed-floor), cannot represent the $\sim 5 \times 10^4$ effective degrees of freedom of the methylome with thousands of inducing points and lands far from the exact GPs (RMSE 0.3804 vs. 0.2448). VNNGP—nominally the right approximate method for one-dimensional local structure—fares even worse (0.4222), for an instructive reason: its sequential nearest-neighbor setup only completed at a 200 000-site subsample, and subsampling a 2.1M-site chromosome to 200k stretches the median inter-site gap several-fold, destroying precisely the local correlation the method relies on. Its tractability ceiling and its accuracy are not independent failures; the first causes the second.

4.3 Where the gains come from

The ablation row isolates the contributions, and the answer is sharper than we anticipated: swapping in the coverage-aware noise and density mean while keeping the stationary kernel moves CRPS only from 0.1368 to 0.1360 and leaves RMSE unchanged (0.2484 vs. 0.2483)—the observation model’s value is concentrated in depth-resolved calibration, not in aggregate scores. The bulk of the gain comes from the *metric*: the warp (with its variance field) accounts for the remaining improvement to 0.1341/0.2448. Per-class breakdowns (Table 2) localize that advantage where density structure is strongest: islands (RMSE 0.1492 vs. 0.1703) and shores (0.2286 vs. 0.2365), with open sea nearly tied (0.2495 vs. 0.2520)—as expected once the site-identity floor of Section 5.2 dominates. Both kernels *over-cover* inside islands (coverage 0.971 and 0.955 at nominal 0.9), ours slightly more; Section 5.3 traces this shared failure to the compartment-independent nugget and marks it as the clearest model improvement this analysis surfaces.

4.4 Computational profile

At the MAP sample the certified bandwidth is 51 (covariance sparsity 0.0041% nonzero), the banded storage is a few GB, and one exact likelihood evaluation—assembly, Cholesky, solve, log-determinant on all 2 123 184 sites—takes seconds; the full 500-update chain runs in 0.73 h at peak memory 15.1 GB (Figure 3 shows the chain diagnostics; per-block acceptance 0.11/0.26/0.14). The stationary

kernel, forced to buy island coherence with a base-pair radius, drives its bandwidth to 340 and its chain to 1.40 h—the sparsity–adequacy bind quantified: in the physical metric one *cannot* afford the support that dense compartments need. The reference fvGP dask pipeline agrees with the engine to relative difference $1.2\text{e-}15$ at full scale while taking 27.1 s per likelihood evaluation versus 0.8 s—the certificate, not the implementation, is the speedup.

5 Analysis

5.1 Mechanism: why the warp captures what coordinates cannot

Figure 1 visualizes the fitted kernel. In warped space the correlation profile is a single Wendland bump of radius $\rho = 46.3$ (posterior mean 46.3); mapped back to base pairs it automatically stretches and compresses with local density—the physical support reaches 1.45 kb at the median island spacing (29 bp between sites) and 4.8 kb at the median open-sea spacing (104 bp), a $3.3\times$ dynamic range—while remaining one two-parameter family. The variance field contributes a milder, consistent modulation ($v_1 = 0.09$: amplitude rises $\approx 25\%$ from sparse to dense compartments; note this describes the *latent residual* field after the density-driven mean absorbs the island/sea level difference—*island boundaries and shores* are where the residual varies most—so it does not contradict the low raw variance of island cores), and the noise model separates read-depth noise from site-level structure with overdispersion $\alpha = 0.50$ and nugget $\tau = 0.230$. A coordinate-parameterized field would need a parameter per island to imitate the warp’s density coupling; the covariate parameterization needs one (γ), because the genome itself supplies the spatial pattern. This is the concrete sense in which the domain *demands* a covariate-driven kernel: the non-stationarity is not free-form, it is slaved to a known, cheap covariate, and the kernel family should say so.

5.2 What the exact posterior says about the methylome

Co-methylation propagates in CpG-index space. The metric-interpolation weight is identified sharply: $\gamma = 0.897$, 95% credible interval $[0.892, 0.902]$ (Figure 6); three independent chromosome-21 chains agree (posterior means 0.810–0.885). The data overwhelmingly prefer distances counted in *sites* over distances counted in *base pairs*—quantitative, chromosome-scale support for the picture in which maintenance-methylation dynamics act per CpG (DNMT1 processivity [Ren et al., 2022]) rather than per unit length, previously argued from kilobase-scale windows and block statistics [Guo et al., 2017, Affinito et al., 2020]. The practical consequence for modelers: a stationary kernel in bp is not merely suboptimal, it is the *wrong metric*, and its failure concentrates exactly where density departs from the mean—*islands* (over-smoothed) and *PMD interiors* (under-reached).

Compartment-resolved predictability. The per-class breakdown turns the posterior into a map of where the methylome is predictable from its neighborhood: island sites are recovered with RMSE 0.1492 (they behave as coherent, quiet units), while open-sea and PMD sites carry a large site-level component (RMSE 0.2495), consistent with the disordered methylation of HepG2’s partially methylated domains [Salhab et al., 2018]. The predictive distributions transfer across replicates almost perfectly: trained on replicate 2 alone, the model scores replicate 1’s observations at the 13 957 matched held-out sites with 90%-interval coverage 0.880 and standardized residuals of standard deviation 1.024 (Figure 6c)—an independent-library validation that the coverage-aware error model is quantitatively right.

The nugget is biology, not noise. The replicate pair also lets us dissect the fitted noise mechanistically (Figure 7). If the nugget $\tau = 0.230$ were measurement noise, squared differences between the two replicates would approach $2\tau^2 = 0.106$ at high depth; instead they fall to 0.0052 at depth ≥ 40 (over 404 625 site pairs)—95.1% of the nugget variance is *reproducible* across independent libraries. Single-CpG deviations from the regional field are therefore predominantly stable site identity (single-site epigenetic state, allele-specific methylation, and some genetically caused calls such as C>T variants that bisulfite sequencing reads as unmethylated), not sampling error. Two consequences follow. First, an imputation limit: *any* neighborhood-based method—GP or otherwise—faces an accuracy floor at these sites, and our exact posterior quantifies that floor genome-wide instead of silently smoothing over it (the depth-resolved test-set curves in Figure 7 confirm the total predictive variance remains calibrated for single-replicate prediction, where stable site deviations are

epistemically noise). Second, a measurement-design implication: at typical depths the read-sampling component that deeper sequencing can remove is comparable to or smaller than the reproducible site component (replicate squared differences fall from 0.035 at depth 5–10 to 0.0052 at ≥ 40), so beyond moderate depth, budget is better spent on breadth than on depth—a conclusion only available because the noise model separates the two components with calibrated uncertainty.

Sparsity is biology here. The discovered sparsity pattern is not an algorithmic convenience, and its shape is itself the metric finding restated: with γ near the index metric, every site couples to a nearly constant ≈ 90 neighbors while the *physical reach* of its support tracks local CpG density across two orders of magnitude (Figure 2)—tight blocks on islands, wide fringes over deserts—so the covariance support literally draws the chromosome’s regulatory architecture, equalizing *information per site* rather than per base pair. Exactness matters for these claims: because solves and log-determinants are computed to machine precision on the full training set, differences between kernels are model differences, and the γ posterior is a statement about the methylome rather than about an approximation’s inductive bias.

5.3 Failure modes and limitations

Honest failures localize the method’s boundary. (i) *Site-level deviations bound point accuracy*: individual CpGs deviate from their neighborhood by $\tau = 0.230$ in fraction units, and Section 5.2 shows those deviations are mostly reproducible biology; consequently RMSE differences between all reasonable smoothers are compressed (the model’s advantage concentrates in the distributional metrics), and no neighborhood-based method should be expected to breach this floor. (i') *Soft identifiability of the noise split*: the overdispersion α was still drifting slowly at chain end ($0.79 \rightarrow 0.50$; Figure 3d) while the total predictive variance stayed stable—the likelihood constrains $\alpha v_i + \tau^2$ tightly but its decomposition weakly, which is why the decomposition is validated against replicates rather than against the chain alone. (i'') *Uniform nugget over-covers quiet compartments*: both kernels’ 90% intervals cover ≈ 0.96 – 0.97 of island sites (nominal 0.90) because the site-identity component is much smaller inside islands than in PMD open sea while τ is compartment-independent; this is the largest remaining calibration defect (it dominates both models’ aggregate ECE: 0.0741 for ours, 0.0657 for the stationary kernel) and points directly at a density-indexed nugget $\tau(c)$ as the next model refinement—identified here, deliberately not retro-fitted, since it was not part of the pre-registered construction. (ii) *Sparsity budget*: the bandwidth budget $B_{\max} = 768$ never bound—zero rejected proposals for every kernel—so the stationary row is its own likelihood optimum, not a budget-truncated one: given all the support it asked for (bandwidth 340, $7\times$ ours, at $5.7\times$ our per-evaluation cost), the bp-metric kernel still trails by 33 954 nats. The budget’s role was preventive (it rejects the pathological wide-support proposals early chains can make), and the cost asymmetry it permits is itself a finding: the wrong metric pays more to fit worse. (iii) *MCMC scale*: 500 exact-likelihood updates suffice for the 10-parameter adapted kernel (stable per-block acceptance 0.11/0.26/0.14, flat log-marginal-likelihood tail) at 0.73 h; the 12-parameter convolution kernel needs $8\times$ the wall-clock for its 500 updates (acceptance 0.14/0.12) because its bp-metric bandwidth makes every exact evaluation expensive—coordinate-expanded fields are not just statistically inadequate here, they are also costly to sample. (iv) *Scope*: one chromosome, one cell line, one assay; chromosome-21 seed replicates and the cross-replicate check probe stability, but pan-genome and pan-tissue generality, CpH methylation, and array platforms remain untested. (v) *Model class*: the Gaussian observation model approximates a binomial; at coverage 5 the approximation is visible in the extreme-level tails of the calibration curve.

6 Conclusion

We modeled a human chromosome’s methylome with an exact Gaussian process at single-CpG resolution—2 123 184 training sites, exact solves, log-determinants and block-MCMC on one CPU node—by designing the kernel the domain asks for: compact support in a learned interpolation between physical and CpG-index metrics, amplitude modulated by sequence-derived CpG density, and read-depth-aware noise. The construction certifies its own sparsity, turning the quadratic assembly of sparsity-discovering GPs into a linear-in- N banded pipeline, and it outperforms the existing compact kernels of that family—which we show are metric-mismatched on this domain, by tens of thousands of nats—as well as best-effort variational baselines. The posterior delivers two crisp biological

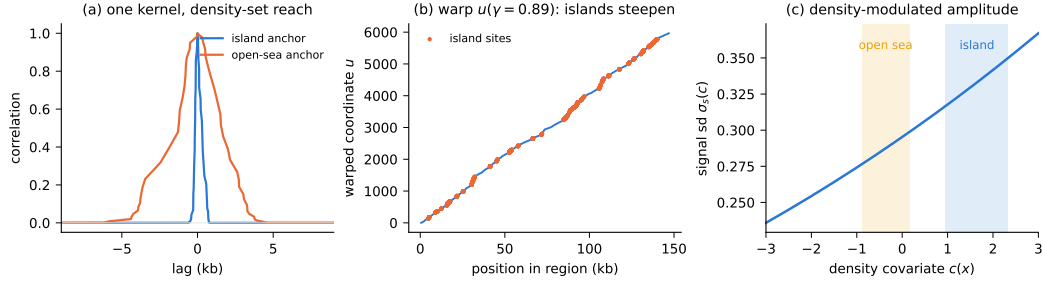


Figure 1: The fitted density-warped kernel. Left: correlation vs. base-pair lag anchored at an island site and at an open-sea site—one warped-space bump, density-dependent physical reach. Middle: warped coordinate $u(\gamma)$ vs. position over a 150 kb island-rich region ($\gamma = 0.894$): islands steepen the warp, deserts flatten it. Right: the signal-sd field $\sigma_s(c)$ over the density covariate, with island and open-sea interquartile bands.

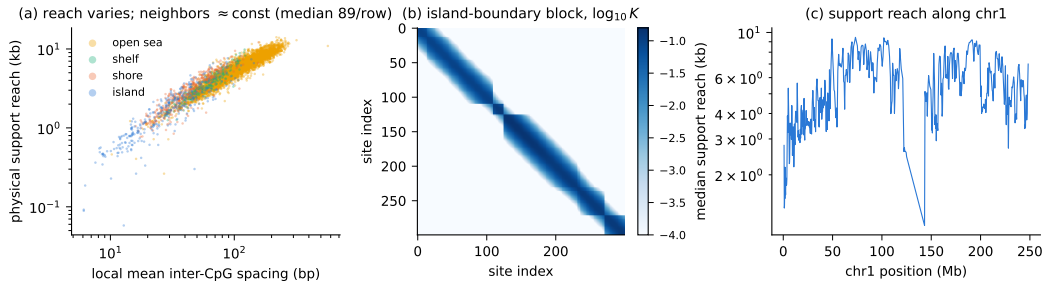


Figure 2: Certified sparsity at the MAP. Left: each row’s physical support reach vs. local mean spacing, colored by compartment—reach spans two orders of magnitude while nonzeros per row stay nearly constant (median 89), the index-metric signature. Middle: banded structure of a 300-site island-boundary block (log-magnitude). Right: median support reach along the chromosome (the centromere gap is visible near 125–145 Mb).

statements: co-methylation is organized per CpG site ($\gamma = 0.897$), and the part of single-CpG variation that no neighborhood can predict is itself mostly reproducible site identity (95.1% across independent libraries), not noise—an imputation floor now quantified with calibrated uncertainty. We see two directions with leverage: joint multi-replicate and multi-sample GPs (the noise model validated across replicates here is the entry point), and active sequencing—using calibrated posteriors to decide where additional depth buys the most information, the decision-theoretic use case that point estimators cannot serve and that exactness at scale now makes available to epigenomics.

References

- O. Affinito et al. Nucleotide distance influences co-methylation between nearby CpG sites. *Genomics*, 112(1):144–150, 2020.
- C. Angermueller, H. J. Lee, W. Reik, and O. Stegle. DeepCpG: accurate prediction of single-cell DNA methylation states using deep learning. *Genome Biology*, 18:67, 2017.
- D. Bolin and J. Wallin. Spatially adaptive covariance tapering. *Spatial Statistics*, 18:163–178, 2016.
- G. De Waele, J. Clauwaert, G. Menschaert, and W. Waegeman. CpG Transformer for imputation of single-cell methylomes. *Bioinformatics*, 38(3):597–603, 2022.
- A. Datta, S. Banerjee, A. O. Finley, and A. E. Gelfand. Hierarchical nearest-neighbor Gaussian process models for large geostatistical datasets. *JASA*, 111(514):800–812, 2016.
- F. Eckhardt et al. DNA methylation profiling of human chromosomes 6, 20 and 22. *Nature Genetics*, 38:1378–1385, 2006.

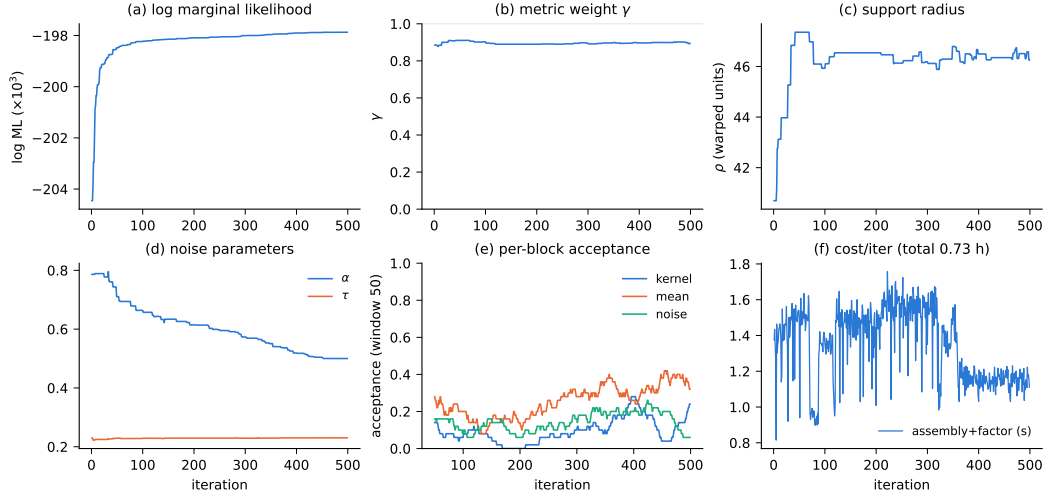


Figure 3: BMH-MCMC diagnostics at $N = 2123184$: log marginal likelihood, γ , ρ , and noise parameters along the chain; per-block acceptance rates; wall-clock per iteration.

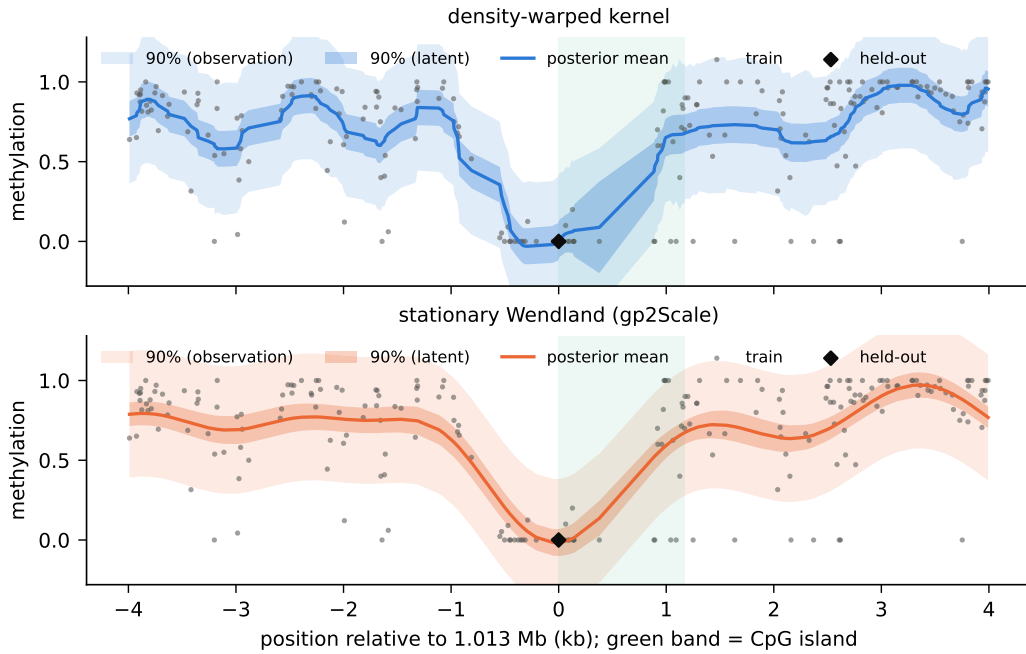


Figure 4: Exact posterior over an island–open-sea boundary (train sites, held-out sites, posterior mean and 90% bands; observation-level and latent-level uncertainty distinguished). The stationary kernel over-smooths the island edge and inflates island uncertainty.

Y. Feng, Z. Zhang, Y. Hong, et al. Methylation haplotype blocks and their landscape in the human genome. *Genome Research*, 33(12):2041–2052, 2023.

M. Frank. *Modeling epigenetic heterogeneity across time and genome in single-cell multi-omics experiments*. PhD thesis, Heidelberg University, 2024.

R. Furrer, M. G. Genton, and D. Nychka. Covariance tapering for interpolation of large spatial datasets. *J. Computational and Graphical Statistics*, 15(3):502–523, 2006.

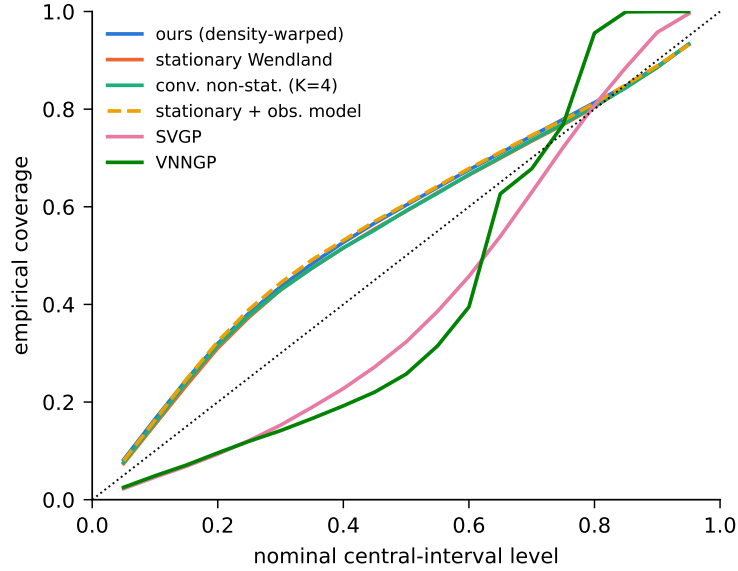


Figure 5: Probability-coverage curves on the frozen test set: empirical vs. nominal central-interval coverage for the adapted kernel, the stationary Wendland, the convolution non-stationary kernel, the ablation, and the variational baselines. Per-compartment and per-depth coverage values are reported in Table 2 and Section 4.2.

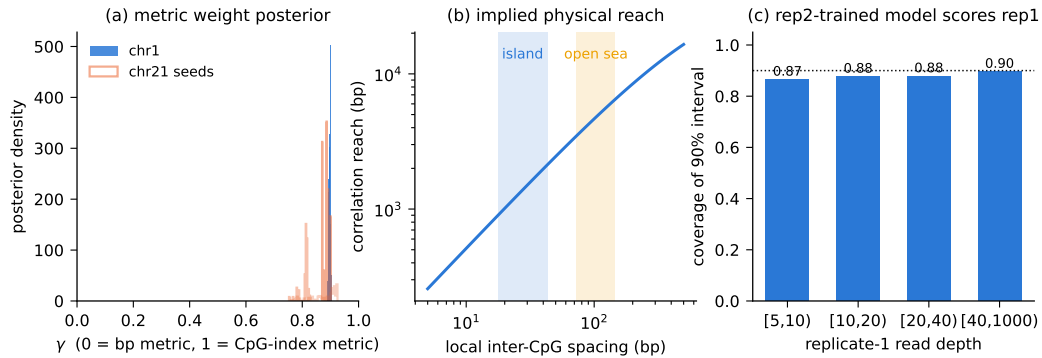


Figure 6: The metric result. Left: posterior of γ (chr1 chain; chr21 seed replicates overlaid). Middle: implied correlation length in bp vs. local spacing with compartment annotations. Right: replicate-2-trained predictive coverage evaluated on replicate 1 (noise-model validation).

M. N. Gibbs. *Bayesian Gaussian processes for regression and classification*. PhD thesis, University of Cambridge, 1997.

T. Gneiting. Compactly supported correlation functions. *J. Multivariate Analysis*, 83(2):493–508, 2002.

T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *JASA*, 102(477):359–378, 2007.

S. Guo et al. Identification of methylation haplotype blocks aids in deconvolution of heterogeneous tissue samples and tumor tissue-of-origin mapping from plasma DNA. *Nature Genetics*, 49:635–642, 2017.

V. Halla-aho and H. Lähdesmäki. LuxUS: DNA methylation analysis using generalized linear mixed model with spatial correlation. *Bioinformatics*, 36(17):4535–4543, 2020.

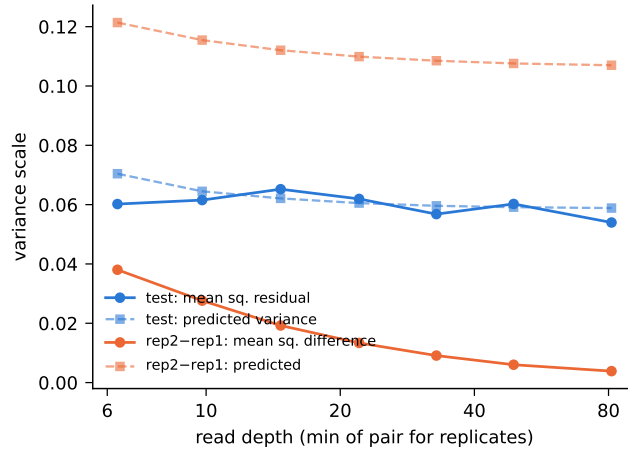


Figure 7: Dissecting the noise model with the replicate pair, by read depth. Blue: on the held-out test set, mean squared residuals match the model’s total predictive variance (single-replicate prediction is calibrated). Orange: squared *differences between replicates* track the depth-dependent read-sampling floor but fall far below the nugget prediction $2\tau^2$ at high depth—95.1% of the site-level variance is reproducible across libraries, i.e. stable site identity rather than noise (Section 5.2).

- K. D. Hansen, B. Langmead, and R. A. Irizarry. BSmooth: from whole genome bisulfite sequencing reads to differentially methylated regions. *Genome Biology*, 13:R83, 2012.
- M. Heinonen, H. Mannerström, J. Rousu, S. Kaski, and H. Lähdesmäki. Non-stationary Gaussian process regression with Hamiltonian Monte Carlo. In *AISTATS*, pages 732–740, 2016.
- J. Hensman, N. Fusi, and N. D. Lawrence. Gaussian processes for big data. In *UAI*, 2013.
- M. Katzfuss and J. Guinness. A general framework for Vecchia approximations of Gaussian processes. *Statistical Science*, 36(1):124–141, 2021.
- C. G. Kaufman, M. J. Schervish, and D. W. Nychka. Covariance tapering for likelihood-based estimation in large spatial data sets. *JASA*, 103(484):1545–1555, 2008.
- K. Korthauer, S. Chakraborty, Y. Benjamini, and R. A. Irizarry. Detection and accurate false discovery rate control of differentially methylated regions from whole genome bisulfite sequencing. *Biostatistics*, 20(3):367–383, 2019.
- C. Lökvist, I. B. Dodd, K. Sneppen, and J. O. Haerter. DNA methylation in human epigenomes depends on local topology of CpG sites. *Nucleic Acids Research*, 44(11):5123–5132, 2016.
- J. Mateu, G. Fernández-Avilés, and J. M. Montero. On a class of non-stationary, compactly supported spatial covariance functions. *Stochastic Environmental Research and Risk Assessment*, 27:297–309, 2013.
- M. M. Noack and J. A. Sethian. Advanced stationary and nonstationary kernel designs for domain-aware Gaussian processes. *Comm. App. Math. and Comp. Sci.*, 17(1):131–156, 2022.
- M. M. Noack, H. Krishnan, M. D. Risser, and K. G. Reyes. Exact Gaussian processes for massive datasets via non-stationary sparsity-discovering kernels. *Scientific Reports*, 13:3155, 2023.
- M. M. Noack, M. D. Risser, H. Luo, V. Tekriwal, and R. J. Pandolfi. gp2Scale: a class of compactly-supported non-stationary kernels and distributed computing for exact Gaussian processes on 10 million data points. arXiv:2512.06143, 2025.
- C. J. Paciorek and M. J. Schervish. Spatial modelling using a new class of nonstationary covariance functions. *Environmetrics*, 17(5):483–506, 2006.

- H. Ren, R. B. Taylor, T. L. Downing, and E. L. Read. Locally correlated kinetics of post-replication DNA methylation reveals processivity and region specificity in DNA methylation maintenance. *J. Royal Society Interface*, 19(195):20220415, 2022.
- M. D. Risser and C. A. Calder. Regression-based covariance functions for nonstationary spatial modeling. *Environmetrics*, 26(4):284–297, 2015.
- M. D. Risser and D. Turek. Bayesian inference for high-dimensional nonstationary Gaussian processes. *J. Statistical Computation and Simulation*, 90(16):2902–2928, 2020.
- M. D. Risser, M. M. Noack, H. Luo, and R. Pandolfi. Compactly-supported nonstationary kernels for computing exact Gaussian processes on big data. *Environmetrics*, 36:e70054, 2025.
- A. Salhab et al. A comprehensive analysis of 195 DNA methylomes reveals shared and cell-specific features of partially methylated domains. *Genome Biology*, 19:150, 2018.
- J. Snoek, K. Swersky, R. Zemel, and R. P. Adams. Input warping for Bayesian optimization of non-stationary functions. In *ICML*, pages 1674–1682, 2014.
- O. Taryma-Leśniak et al. Methylation patterns at the adjacent CpG sites within enhancers are a part of cell identity. *Epigenetics & Chromatin*, 17, 2024.
- A. Taudt et al. METHimpute: imputation-guided construction of complete methylomes from WGBS data. *BMC Genomics*, 19:444, 2018.
- A. V. Vecchia. Estimation and model identification for continuous spatial processes. *JRSS B*, 50(2):297–312, 1988.
- K. Wang, G. Pleiss, J. Gardner, S. Tyree, K. Q. Weinberger, and A. G. Wilson. Exact Gaussian processes on a million data points. In *NeurIPS*, 2019.
- H. Wendland. Piecewise polynomial, positive definite and compactly supported radial functions of minimal degree. *Advances in Computational Mathematics*, 4:389–396, 1995.
- L. Wu, G. Pleiss, and J. P. Cunningham. Variational nearest neighbor Gaussian process. In *ICML*, pages 24114–24130, 2022.
- W. Zhang, T. D. Spector, P. Deloukas, J. T. Bell, and B. E. Engelhardt. Predicting genome-wide DNA methylation using methylation marks, genomic position, and DNA regulatory elements. *Genome Biology*, 16:14, 2015.
- K. Zhao et al. A novel statistical method for modeling covariate effects in bisulfite sequencing derived measures of DNA methylation. *Biometrics*, 77(2):424–438, 2021.
- L. S. Zou et al. BoostMe accurately predicts DNA methylation values in whole-genome bisulfite sequencing of multiple human tissues. *BMC Genomics*, 19:390, 2018.